

# NextGenC — Gridded SOC maps of Finland

Procedural note for the kriged SOC map products that accompany the NextGenC report. Companion to `build_soc_matrices.R` (point matrices) and `build_soc_maps.R` (this product).

## 1 What this produces

For each of the 6 models (SP1, TP2, TP3, Yasso07, Yasso15, Yasso20), the posterior-predictive SOC at the  $\sim 447$  calibration plots is interpolated to a regular grid covering Finland, **one layer per year, 1985–2024 (40 years)**. Two multiband GeoTIFFs per model, plus a multi-model ensemble pair (**14 files total**):

| File                                    | Bands | Content                                                         |
|-----------------------------------------|-------|-----------------------------------------------------------------|
| <code>&lt;MODEL&gt;_SOC_mean.tif</code> | 40    | Kriged mean total SOC (tC/ha), band $n = \text{year } 1984 + n$ |
| <code>&lt;MODEL&gt;_SOC_sd.tif</code>   | 40    | Total uncertainty SD (tC/ha), same grid/year as mean band       |
| <code>ENSEMBLE_SOC_mean.tif</code>      | 40    | Six-model ensemble mean SOC (tC/ha)                             |
| <code>ENSEMBLE_SOC_sd.tif</code>        | 40    | Ensemble total SD (within + between-model + spatial)            |

Band names carry the year (e.g. `y1985`), written into GeoTIFF metadata so labels travel with the file. Band  $n$  of every SD stack is the uncertainty for band  $n$  of the matching mean stack (same year, grid, and land mask).

## 2 Grid and storage

- **CRS:** ETRS-TM35FIN (EPSG:3067), metres.
- **Resolution:** 2 km. Grid  $360 \times 600$  ( $\sim 216$  k cells, of which  $\sim 42\%$  land).
- **Mask:** clipped to the Finland boundary from `maakunta_sf.rds`.
- **Format:** GeoTIFF, DEFLATE-compressed, float32.
- **Footprint:**  $\sim 0.12$  GB for all 560 maps (14 files).

The 2 km choice is a deliberate compromise: the  $\sim 447$  plots have a **mean spacing of  $\sim 27.5$  km**, so 2 km already oversamples the information content — finer grids render smoother but add no real spatial detail. Disk budget (1–2 GB) is not the binding constraint; data density is.

## 3 Method

Kriging is done **in log space** and back-transformed, because SOC is positive and right-skewed and HIKET’s error model is multiplicative — this keeps the map uncertainty multiplicative and consistent with the calibration. Per model  $\times$  year:

1. **Inputs.** Plot mean and SD come from the posterior-predictive bundles (the same `soc_mean` / `soc_sd` columns used by `build_soc_matrices.R`); plot coordinates (`x_ETRS`, `y_ETRS`) come from `Data/model_inputs/site`, joined on `plot_id`.
2. **Krige the mean.** Ordinary kriging of `log(soc_mean)` over the grid  $\rightarrow$  interpolated mean surface plus the **kriging prediction variance**  $v_{\text{krige}}$  (small near plots, large in data-sparse areas, e.g. Lapland).
3. **Krige the model uncertainty.** Ordinary kriging of `log(soc_sd)`  $\rightarrow$  a surface of the model’s own posterior-predictive uncertainty.
4. **Combine (option C).** Total uncertainty fuses both sources,

$$\text{sd}_{\text{total}} = \sqrt{v_{\text{krige}} + \text{kriged\_modelSD}^2}$$

(combined in log space, then back-transformed). This propagates **model uncertainty** and **spatial-coverage uncertainty** into a single SD layer — neither source is silently dropped.

5. **Back-transform & mask.** Exponentiate to tC/ha, clip to the land mask, write the two stacks.

Variogram: **one pooled model per model**, reused for all 40 years (the spatial structure of SOC is essentially static year-to-year; this is stable and avoids per-year auto-fit failures). `set.seed(2025)` as elsewhere in HIKET.

## 4 Ensemble (six-model) product

The ensemble is formed *on the grid*, after each model has been kriged — i.e. the per-model mean and total-SD stacks above are combined cell-by-cell, equal-weighted ( $w_i = 1/6$ ). With  $M = 6$  model mean surfaces  $m_i$  and total-SD surfaces  $s_i$  (each  $s_i$  already folds in that model's kriging variance and posterior uncertainty, §3 step 4):

$$m_{\text{ens}} = \frac{1}{M} \sum_i m_i, \quad s_{\text{ens}} = \sqrt{\underbrace{\frac{1}{M} \sum_i s_i^2}_{\text{within-model}} + \underbrace{\frac{1}{M-1} \sum_i (m_i - m_{\text{ens}})^2}_{\text{between-model (structural)}}}.$$

This is the law of total variance: the ensemble deviation fuses **(i)** each model's own predictive uncertainty (which already carries **(iii)** the spatial/kriging uncertainty) with **(ii)** the *structural spread* between models — the among-model disagreement that is the central HIKET question. The between-model term is the scientifically interesting one: where it dominates, the map is uncertain because the models *disagree*, not because the data are sparse. Combination is done in the same (log/back-transformed) space as the per-model SDs for consistency.

**Note:** the between-model term is also informative on its own; if a structural-disagreement map is wanted later, export  $\sqrt{\frac{1}{M-1} \sum_i (m_i - m_{\text{ens}})^2}$  as a separate stack — one extra line in the script.

## 5 How to run

# from the project root

Rscript Reporting/NextgenC\_report/build\_soc\_maps.R

Reads the same bundles `<- c(...)` `RUN_ID` map as `build_soc_matrices.R`. Writes the 12 GeoTIFFs into `Reporting/NextgenC_report/`.

**Dependencies:** `terra`, `gstat`, `sf` (and `sp`). Posterior-predictive bundles must be present in `Calibration_real_data_transient/runs/`, and `Data/model_inputs/{site_raw.csv, maakunta_sf.rds}` must exist.

## 6 Maintenance notes

- **TP3 RUN\_ID is provisional.** As in `build_soc_matrices.R`, the TP3 bundle is the pre-fix (Euler) predictive. Swap the TP3 RUN\_ID to the exact-integrator re-calibration when it is synced from Puhti, then re-run.
- **Adding years / plots.** No code change — the script reads year columns and plot rows from the bundles directly.
- **Changing resolution.** Edit the single `RES_M <- 2000` constant. Storage scales as  $1/\text{res}^2$ : 1 km  $\approx$  0.4 GB, 5 km  $\approx$  0.02 GB (all 560 maps, compressed).
- **Reading the output.** `terra::rast("SP1_SOC_mean.tif")` gives a 40-layer `SpatRaster`; layer names are the years. Same for the `.sd` file.